

An Open-Source AES-128 Crypto Accelerator SoC with Design-for-Test: RTL to Signoff-Clean GDSII in SkyWater 130 nm

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Abstract—This paper presents an AES-128 encryption accelerator System-on-Chip (SoC) carried from Verilog RTL to signoff-clean GDSII entirely with open-source tools on the SkyWater 130 nm PDK, with design-for-test (DFT) as a first-class objective. An iterative ten-round datapath with on-the-fly key expansion is integrated behind a UART command interface. Two flow configurations are reported: a 67.6% core-utilization baseline, and a DFT configuration in which all 745 flip-flops are replaced by scan flip-flops (100% sequential scan coverage, zero RTL changes) by a custom OpenLane 2 step operating on the synthesized netlist, alongside a March C⁻ memory BIST verified by fault injection. Both configurations close timing at 50 MHz with zero setup and hold violations across nine corners, pass DRC with zero violations, and match LVS exactly. Encryption latency of 10 cycles (200 ns, 0.64 Gbps) is measured by an in-design cycle counter on an Artix-7 FPGA and independently reproduced in simulation; four known-answer vectors, including the NIST SP 800-38A and FIPS-197 Appendix B references, pass with zero bit errors. Post-layout power analysis on extracted parasitics yields 6.08 mW at the typical corner (0.122 mW/MHz; 1.22 nJ per block). All sources, scripts, and run artifacts are released for reproduction.

Index Terms—AES-128, design-for-test, scan, memory BIST, SkyWater 130nm, OpenLane, open-source EDA, physical design, GDSII, FPGA validation

I. INTRODUCTION

The Advanced Encryption Standard (AES) [1] remains the workhorse of symmetric cryptography in embedded and IoT systems. Two trends make an open, fully documented ASIC implementation valuable today. First, hardware accelerators dominate software on energy-constrained devices: the best published bitsliced AES-128 on an ARM Cortex-M4 requires about 80 cycles/byte [7], whereas the accelerator in this paper completes a 128-bit block in 10 clock cycles. Second, the maturing of open-source EDA flows [10]–[12] and the open SkyWater 130 nm PDK [13] have made full RTL-to-GDSII implementation accessible outside industry; yet published cryptographic implementations that document complete signoff, DFT structures, measured latency, and post-layout power remain scarce.

Most published AES hardware stops at RTL simulation or FPGA prototyping; open-source silicon efforts such as the multi-algorithm crypto accelerator taped out through the Google/SkyWater shuttle program [9] report no measured benchmark data and no DFT. This paper addresses that gap

with a complete, reproducible implementation whose distinguishing property is that every reported number traces to a committed artifact.

Contributions.

- 1) A complete RTL-to-GDSII implementation of an AES-128 SoC in SkyWater 130 nm using OpenLane 2.3.10, with DRC-clean layout, exact LVS match, and zero setup/hold violations across nine timing corners in two flow configurations.
- 2) DFT insertion performed at the netlist level inside an open flow: a custom OpenLane step maps all 745 flip-flops to scan flip-flops (100% sequential coverage, single clock domain) with no RTL modification, and a March C⁻ memory BIST is verified by behavioral fault injection.
- 3) Multi-level validation: four known-answer vectors in simulation including NIST SP 800-38A and FIPS-197 Appendix B; on-FPGA measurement of the 200 ns encryption latency on an Artix-7 FPGA; and disclosed post-layout power analysis on extracted parasitics.
- 4) Documented negative results where tools fell short: scan-chain stitching was attempted at two flow points and is characterized by its error signature; a floorplan sizing pitfall (ignored absolute dimensions) and its fix are reported for the community.

II. BACKGROUND AND RELATED WORK

AES-128 operates on 128-bit blocks under a 128-bit key through an initial key addition and ten rounds of SubBytes, ShiftRows, MixColumns, and AddRoundKey, with the final round omitting MixColumns [1], [2]. Hardware implementations span the design space from compact iterative cores to unrolled pipelines: Satoh et al. established the classic area/throughput trade-off [3]; Mathew et al. demonstrated aggressive low-voltage operation down to 340 mV in 22 nm tri-gate CMOS [4]; and Hammad et al. surveyed energy-efficient IoT implementations [5]. On the software side, Schwabe and Stoffelen reported 554.4 cycles/block for table-based AES-128-CTR on Cortex-M4 [6], and fixslicing currently holds the Cortex-M4 record near 80 cycles/byte [7], while measured mbedTLS runs at 71 cycles/byte [8].

Table I positions this work. No prior open AES SoC publication combines (i) full physical signoff data, (ii) measured on-FPGA latency, (iii) DFT evidence (BIST plus scan-cell mapping), and (iv) post-layout power analysis.

TABLE I
POSITIONING RELATIVE TO REPRESENTATIVE PRIOR WORK

Work	Node	Signoff	DFT	Measured
Sato et al. [3]	0.11 μm	✓	–	✓
Mathew et al. [4]	22 nm	✓	–	✓
Hammad et al. [5]	180/65 nm	partial	–	✓
Singhali [9]	sky130	taped out	–	–
Software: fixslicing [7]	Cortex-M4	n/a	n/a	80 c/B
This work	sky130	✓	✓	✓

III. SoC ARCHITECTURE

A. Iterative AES-128 Core

The core implements one encryption round as combinational logic and reuses it ten times under finite-state-machine control: a 128-bit state register; sixteen parallel combinational S-boxes; wiring-only ShiftRows; GF(2⁸) MixColumns over the AES irreducible polynomial; and an on-the-fly key expansion unit, avoiding storage of eleven round keys. The interface is `clk`, active-low `rst_n`, `start`, 128-bit key and plaintext, `ciphertext`, `done`, and `busy`. The busy window is exactly ten cycles, consistent with the measured latency reported in Section VI.

B. UART SoC Wrapper

The core is wrapped in a minimal SoC with a 115200-baud 8N1 UART interface. A command frame is 33 bytes: command byte 0xAE, 16 key bytes, and 16 plaintext bytes, most-significant byte first. The response frame is 17 bytes: status 0xAA followed by the 16 ciphertext bytes. The final (least significant) ciphertext byte drives eight LEDs, giving a visible known-answer check: 0x97 for the NIST SP 800-38A vector 1. A six-state Moore FSM sequences idle, key load, plaintext load, encryption wait, status transmit, and ciphertext transmit; a non-encrypt command byte asserts an error flag by design. The UART receiver uses a two-flop synchronizer and a 434-clock-per-bit oversampling divider at 50 MHz.

IV. OPEN-SOURCE PHYSICAL DESIGN FLOW

A. Toolchain

The flow is OpenLane 2.3.10 on Ubuntu 22.04: Yosys 0.46 synthesis, OpenROAD (build `edf00dff`) for floorplanning through detailed routing, OpenSTA for timing, Magic 8.3.489 for DRC and extraction, Netgen 1.5.278 for LVS, and KLayout for streaming, targeting the sky130A PDK (volare version `0fe599b2`) with the sky130_fd_sc_hd cell library. All versions are quoted from tool-log banners rather than assumed.

B. Floorplan Sizing: A Documented Pitfall

An early configuration specified absolute die and core rectangles; the flow silently ignored them, fixing utilization at 20.2%. Switching to relative sizing (a core-utilization target with the flow computing dimensions) produced the 67.6% baseline configuration. Because this behavior differs across OpenLane versions, we report it explicitly: absolute sizing keys may be accepted but not honored, and utilization should be verified from placement logs rather than assumed from configuration inputs.

C. DFT-Enabled Configuration

Scan-cell replacement is performed by OpenROAD’s `scan_replace` on the synthesized netlist, inserted into the flow as a custom step (`AesDFT.ScanReplace`) immediately after synthesis and its checkers, before power-connection setup and floorplanning. No RTL is modified at any point. All 745 flip-flops map to scan flip-flops: 731 `sdftrtp_2` (asynchronous reset) and 14 `sdfsbp_2` (asynchronous set, dual output) — 100% sequential coverage in a single clock domain. Because scan cells are larger than their functional counterparts, synthesized cell area grows 44% (116.8k to 168.0 μm^2 at the synthesis stage); the floorplan utilization target is compensated (38% pre-scan-equivalent) so that the placed design lands at a routable 51.9%.

V. DESIGN-FOR-TEST INFRASTRUCTURE

Memory BIST. The 256×8 SRAM is wrapped by a March C[−] controller with a two-wait-state read pipeline matched to the memory’s registered output. Correctness is verified in simulation by behavioral fault injection of both polarities: a fault-free memory passes, and an injected stuck-at-0 at address 42, bit 0 is detected with the failing address reported. This establishes that the BIST detects real defects, not merely that it executes.

Scan-cell mapping. As described in Section IV, the complete sequential population (745/745) is replaced by scan-capable cells inside the flow, yielding a scan-ready netlist and GDS with 100% sequential coverage.

Chain stitching: attempted and characterized. OpenROAD’s `insert_dft` was invoked at two flow points: after detailed placement, and again after post-CTS timing repair (i.e., with a synthesized clock tree present). In both cases the command completes without error but creates zero scan cells, emitting DFT-0005 (“no valid clock connected”) for every flip-flop; 745×2 scan pins consequently remain unconnected in the released GDS, which we disclose. The result is a characterization of the available build: chain stitching with this OpenROAD revision requires clock-domain resolution that the bundled `dft` module does not provide for this design at either insertion point; the scan-ready architecture is the contribution, and stitching with a newer build is future work. Formal equivalence checking is likewise not applicable to an unstitched scan netlist; equivalence is established by simulation (Section VI).

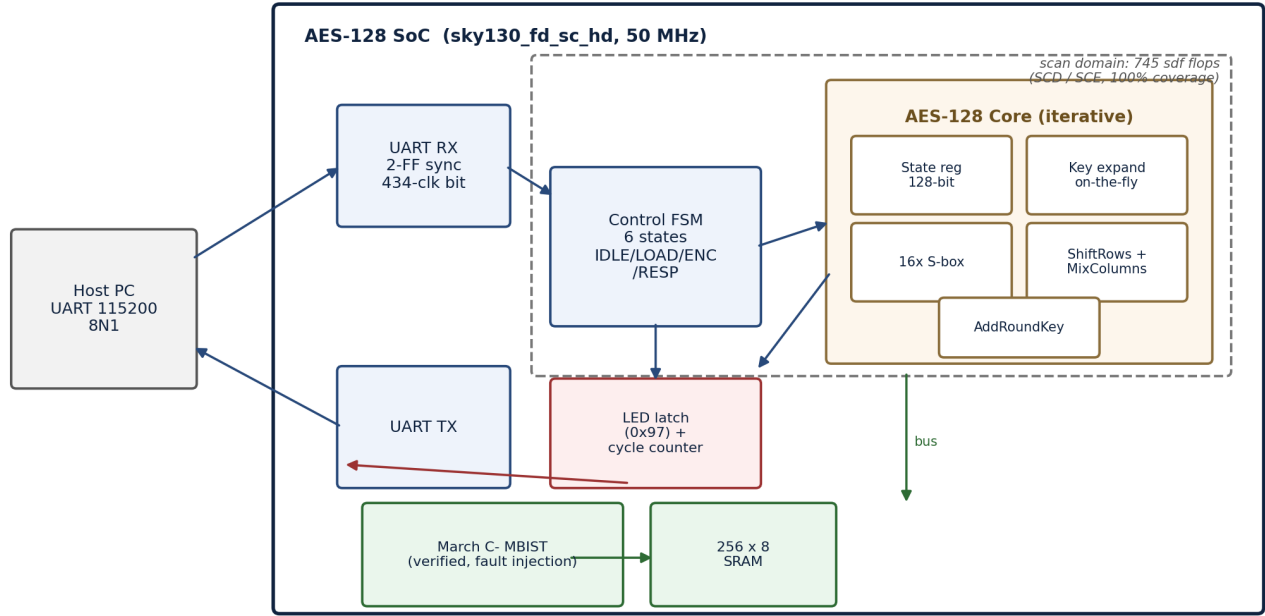


Fig. 1. SoC architecture. The iterative AES-128 core and the six-state control FSM form a single-clock scan domain of 745 scan flip-flops (dashed boundary, SCD/SCE pins); the March C⁻ MBIST verifies the 256×8 SRAM independently of the scan domain.

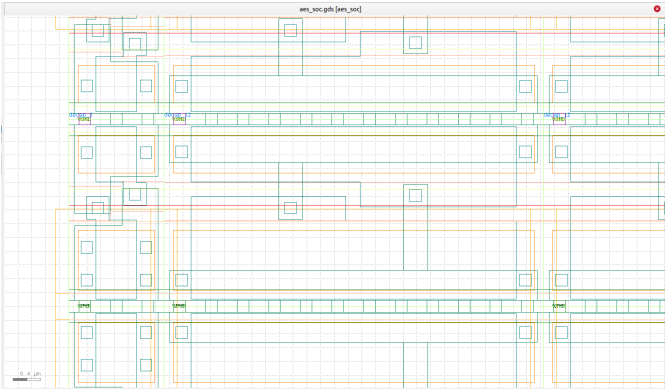


Fig. 2. Placement detail: sky130_fd_sc_hd standard cells (KLayout view).

VI. VERIFICATION, VALIDATION, AND POWER ANALYSIS

A. Functional Verification

RTL simulation on Icarus Verilog (versions 11.0 and 12.0) passes four known-answer vectors: NIST SP 800-38A ECB-AES128 vector 1, FIPS-197 Appendix B, an all-zero KAT, and an all-ones KAT, with zero bit errors. The SoC-level testbench exercises the complete UART protocol—command decode, 33-byte frame reception, status and ciphertext transmission—and checks all 16 ciphertext bytes, the status byte, and the LED value. Two testbench engineering points proved essential and are documented for reuse: pulse-shaped signals (the 10-cycle busy window) must be captured by background monitors rather than sequential waits, and a software UART receiver must re-arm its start-bit edge sensitivity inside the transmitter’s bit

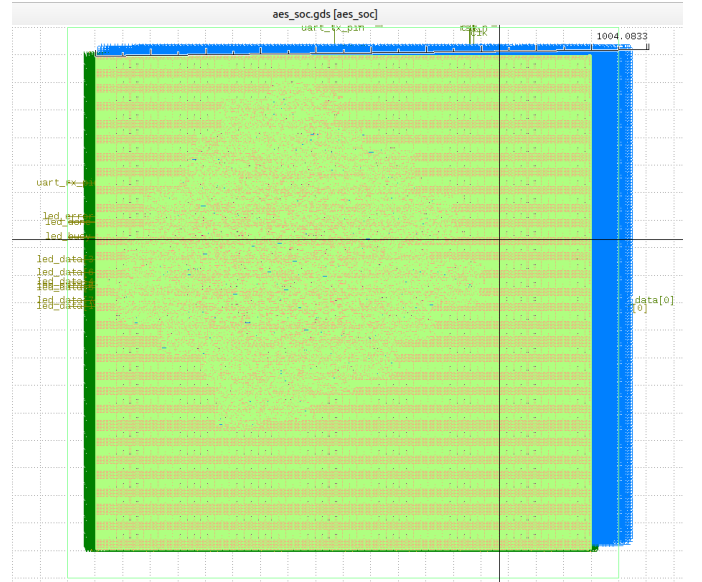


Fig. 3. Full-chip layout view of the open-source flow implementation (KLayout, committed GDSII of the repository).

period (this design uses 435-clock bits with a ~ 2 -clock inter-byte gap), otherwise capture misframes deterministically.

B. FPGA Hardware Validation

The design runs on a Digilent Basys3 board (Artix-7 XC7A35T) under Vivado 2017.4 (WebPACK). A host script sends the SP 800-38A vector-1 frame over USB-UART; the board returns status 0xAA followed by the exact 16-byte NIST

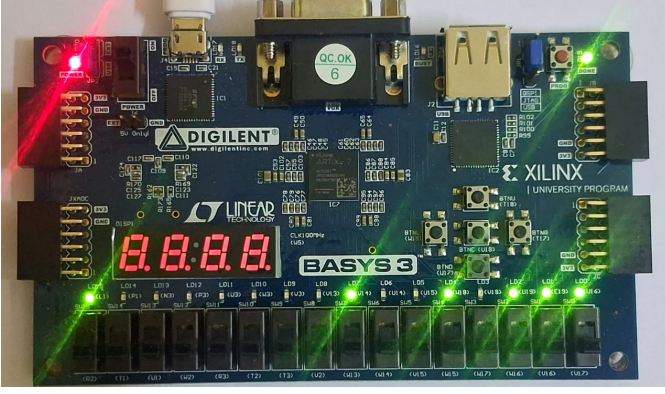


Fig. 4. FPGA validation on a Digilent Basys3 (Artix-7 XC7A35T): LEDs display 0x97, the final ciphertext byte of NIST SP 800-38A vector 1.

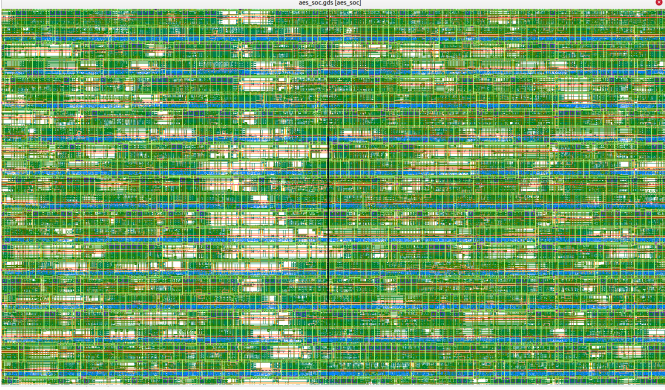


Fig. 5. Metal routing view over the core area (KLayout).

ciphertext, and the LEDs display 0x97. An in-design cycle counter latches the busy window at exactly 10 cycles, i.e., 200 ns at the 50 MHz board clock—agreeing with simulation and establishing the 0.64 Gbps throughput figure by measurement rather than estimation.

C. Post-Layout Power Analysis

Power is *analyzed*, not silicon-measured, and the method is disclosed: OpenSTA on the routed DEF with extracted SPEF parasitics, 50 MHz clock, switching activity 0.10, duty 0.5. Table II reports all three corners. The group split is 39% sequential, 29% combinational, and 31% clock, and the typical-corner total corresponds to 0.122 mW/MHz and 1.22 nJ per 128-bit block.

TABLE II
POST-LAYOUT POWER ANALYSIS (ACTIVITY 0.10, DUTY 0.5, 50 MHz)

Corner	Internal	Switching	Leakage	Total
tt (25 C, 1.80 V)	3.58 mW	2.50 mW	0.05 μ W	6.08 mW
ss (100 C, 1.60 V)	2.91 mW	1.97 mW	55.7 μ W	4.94 mW
ff (−40 C, 1.95 V)	4.07 mW	2.93 mW	0.09 μ W	7.00 mW

VII. RESULTS AND COMPARISON

Table III summarizes both flow configurations; both completed all flow stages to GDSII. Detailed routing converges to zero violations in seven iterations (16062→9323→8993→1312→80→13→0); total wirelength is 560,956 μ m; worst IR drop is 0.021%. Nine timing corners report zero setup and hold violations in both runs.

Against software, the measured 10-cycle block latency compares with 1136 cycles/block for measured mbedTLS AES-128 on Cortex-M4 (71 cycles/byte [8]) — a $114\times$ cycle-count reduction — and with about 1280 cycles/block for the fastest published bitsliced implementation [7] — $128\times$. At equal clock frequency the throughput advantage is of the same order, versus a software baseline that itself outperforms classic table-based code by roughly $2.3\times$ [6].

TABLE III
IMPLEMENTATION SUMMARY (ALL VALUES FROM RUN ARTIFACTS)

	Baseline	DFT-enabled
Die / core (μm^2)	240k / 224k	326k / 308k
Core utilization	67.6%	51.9%
Sequential elements	745	745 (100% scan)
Setup slack (worst/tt)	— / +14.07 ns	+7.22 / +14.06 ns
Hold slack (worst)	+0.29 ns	+0.107 ns
Timing violations (9 corners)	0	0
DRC (Magic / KLayout)	0 / 0	0 / 0
LVS	match	match
Antenna residuals	2 marginal	1 net + 1 pin
MBIST	March C [−] , fault-inj. verified	
Latency @ 50 MHz	10 cycles = 200 ns (FPGA)	
Throughput	0.64 Gbps	
Power (tt, analysis)	—	6.08 mW / 0.122 mW/MHz

VIII. LIMITATIONS AND FUTURE WORK

We state limitations explicitly. (i) The GDS is not fabricated: all physical results are signoff analyses; the FPGA measurement validates function and latency, not silicon power or yield. (ii) Power figures are analyses under a disclosed default activity, not measurements. (iii) Scan chains are mapped but not stitched, as characterized in Section V; 109 nets also lack RC annotation in extraction. (iv) Encryption only: decryption and AES-192/256 are future work. Planned next steps: chain stitching with a newer OpenROAD build; activity-annotated power from gate-level simulation; DPA countermeasures; and silicon measurement through an open-shuttle program.

IX. CONCLUSION

An AES-128 SoC has been implemented from RTL to signoff-clean GDSII in SkyWater 130 nm with fully open tools, with DFT integrated at the netlist level: 100% scan-cell mapping of all 745 flip-flops via a custom flow step, and a fault-injection-verified March C[−] MBIST. The 200 ns measured latency, nine-corner clean timing, zero DRC violations, exact LVS match, and disclosed power analysis—each traceable to a committed artifact—provide a reproducible reference point for open-source cryptographic silicon.

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